

Customer Data

Exploratory Data Analysis Report

Metric	Value
Total Customers	6,418
Churned Customers	1,732
Overall Churn Rate	27.0%
Avg Monthly Charge	\$63.65
Avg Tenure	17.3 months
Total Revenue	\$19,471,390

1. Dataset Overview

The dataset contains **6,418 customer records** across **32 attributes**. It covers demographics, service subscriptions, billing information, and customer status. Of all customers, **1,732 (27.0%)** have churned, **4,275** stayed, and **411** are newly joined.

1.1 Customer Status Distribution

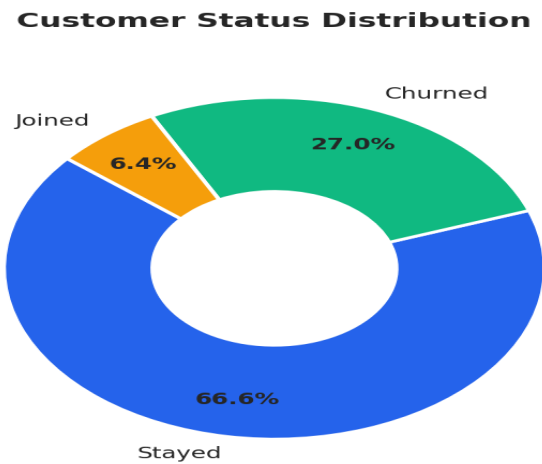


Figure 1: Customer Status Distribution (Donut Chart)

1.2 Descriptive Statistics

Column	Mean	Std Dev	Min	Median	Max
Age	47.1	16.7	18.0	46.0	85.0
Tenure_in_Months	17.3	10.6	1.0	16.0	36.0
Monthly_Charge	63.7	31.2	-10.0	70.1	118.8
Total_Revenue	3033.9	2866.5	21.4	2108.6	11979.3

2. Churn Analysis

Churn analysis is the core focus of this report. The overall churn rate stands at **27.0%**. The dominant churn driver is **Competitor-related reasons** (761 customers), followed by **Attitude** and **Dissatisfaction**.

2.1 Churn by Category

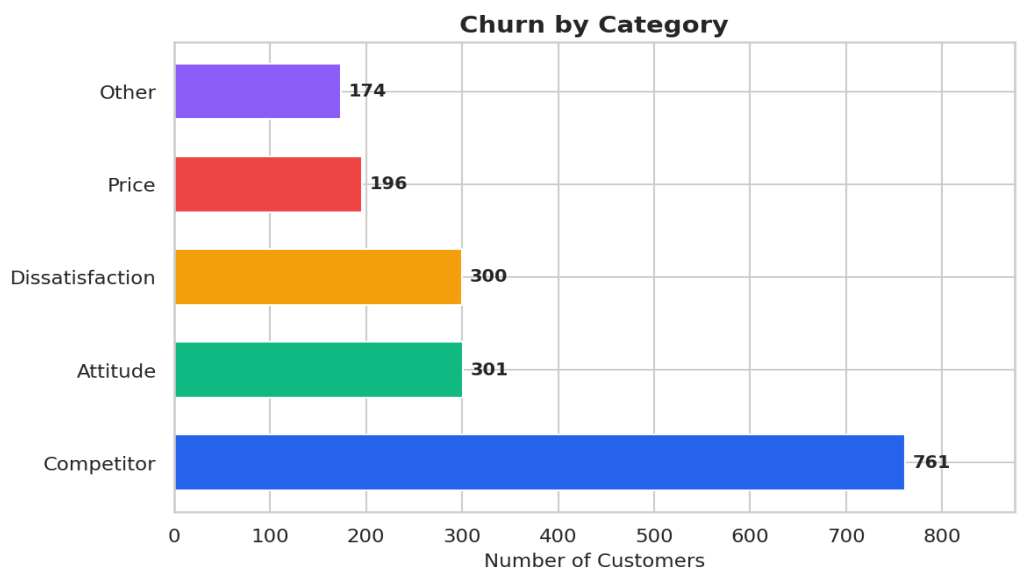


Figure 2: Churn Reasons by Category

2.2 Top 10 Specific Churn Reasons

Rank	Churn Reason	Count	% of Churned
1	Competitor had better devices	289	16.7%
2	Competitor made better offer	274	15.8%
3	Attitude of support person	208	12.0%
4	Don't know	124	7.2%
5	Competitor offered more data	106	6.1%
6	Attitude of service provider	93	5.4%
7	Competitor offered higher download speeds	92	5.3%
8	Price too high	72	4.2%
9	Product dissatisfaction	71	4.1%
10	Network reliability	66	3.8%

2.3 Top 10 Churn Reasons – Chart

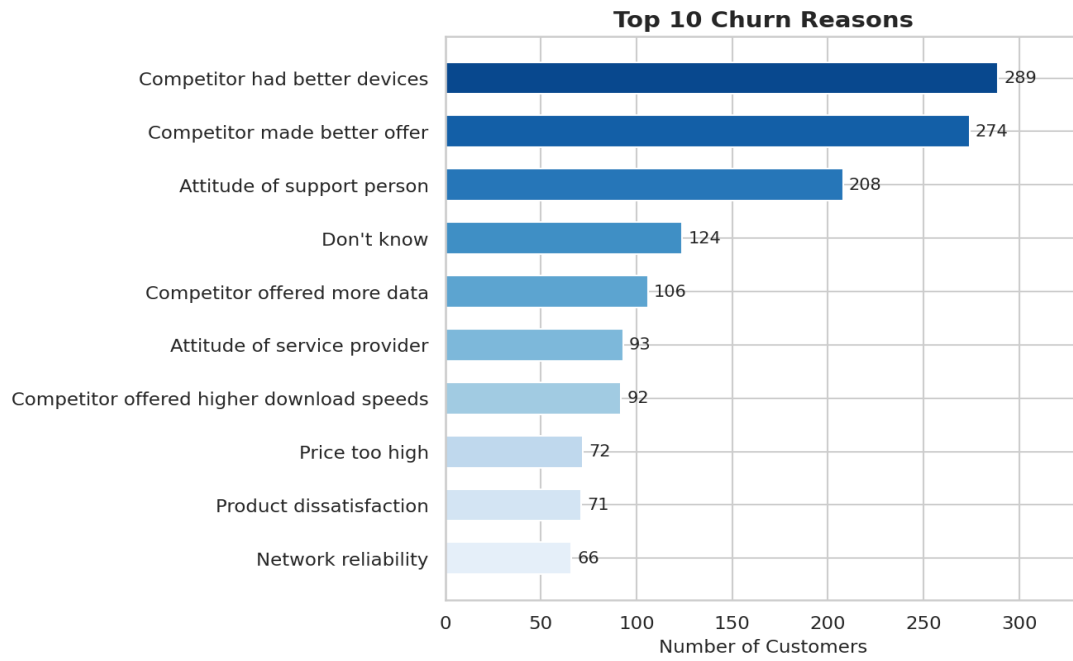


Figure 3: Top 10 Churn Reasons

3. Demographic Analysis

The customer base spans ages **18 to 85** with a mean of **47.1 years**. Churned customers tend to be slightly older on average (50.1 years) compared to those who stayed (46.1 years).

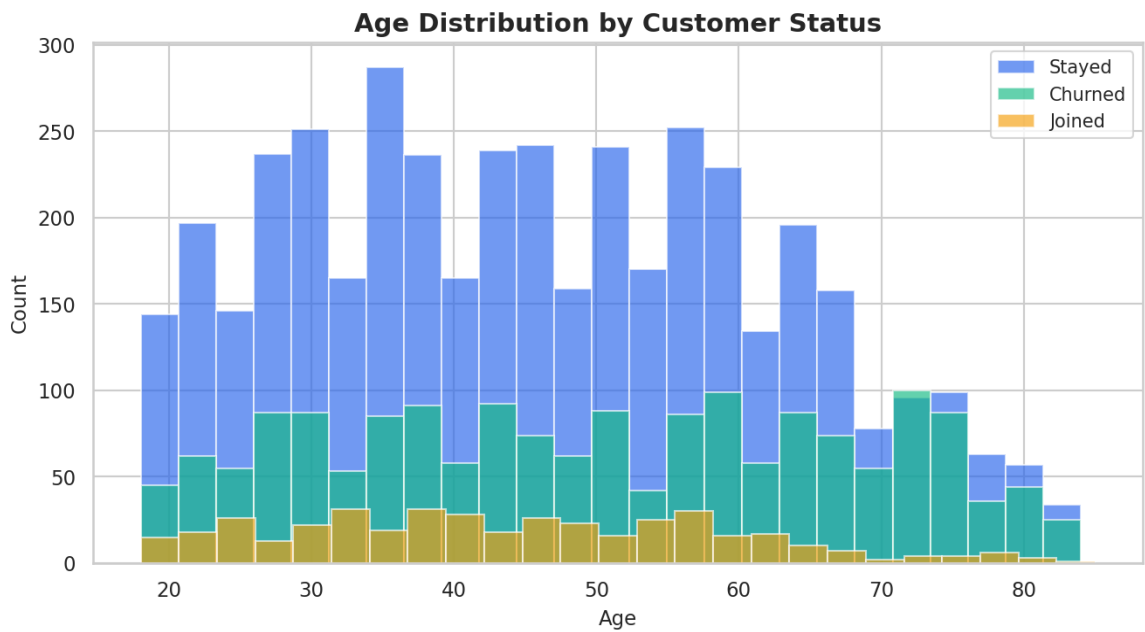


Figure 4: Age Distribution by Customer Status

3.1 Gender Split

Gender	Count	Churn Count	Churn Rate
Male	2,370	621	26.2%
Female	4,048	1,111	27.4%

4. Service & Contract Analysis

4.1 Contract Type vs Churn

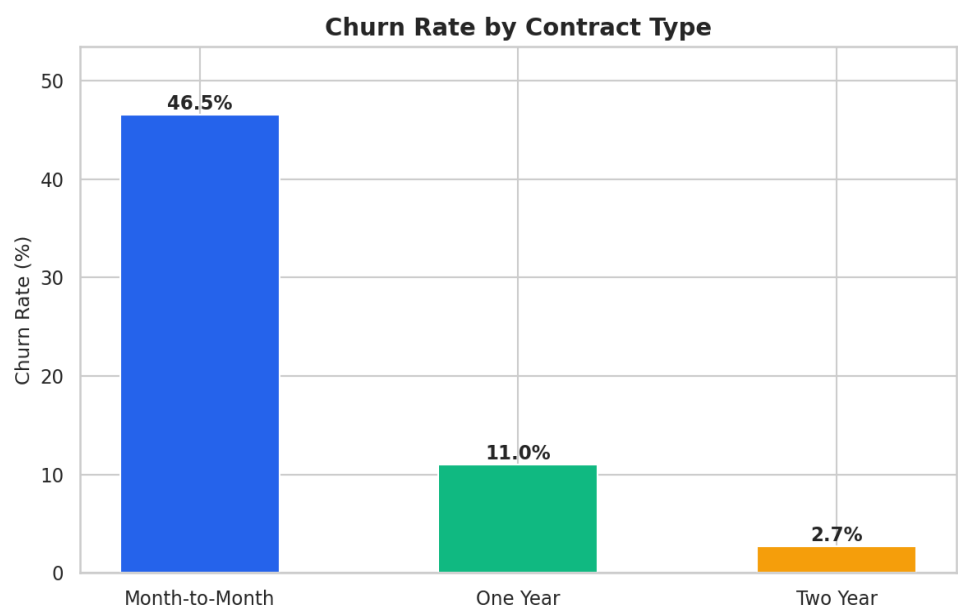


Figure 5: Churn Rate by Contract Type

Month-to-Month contracts have the highest churn rate, indicating that longer-term contracts are strongly associated with customer retention.

4.2 Internet Type vs Churn

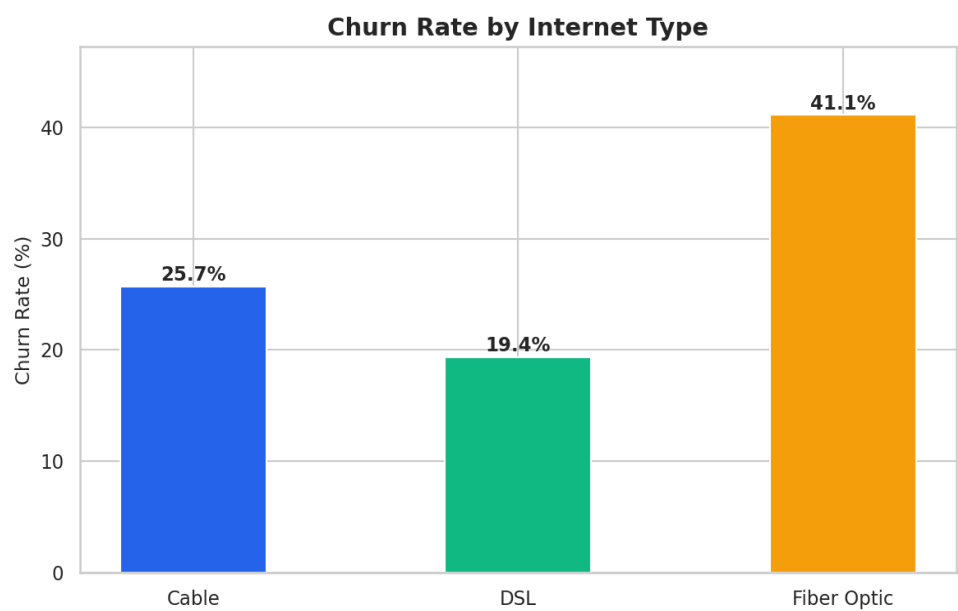


Figure 6: Churn Rate by Internet Type

5. Revenue & Billing Analysis

5.1 Monthly Charge vs Tenure

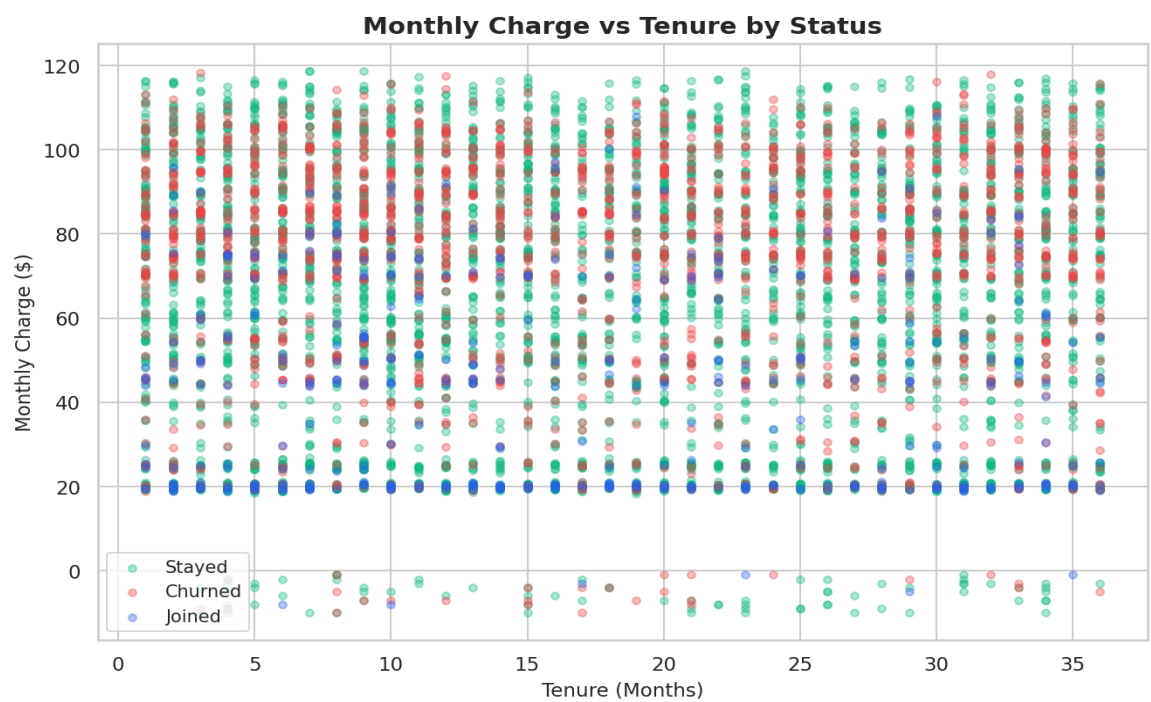


Figure 7: Monthly Charge vs Tenure by Status

5.2 Revenue Distribution by Status

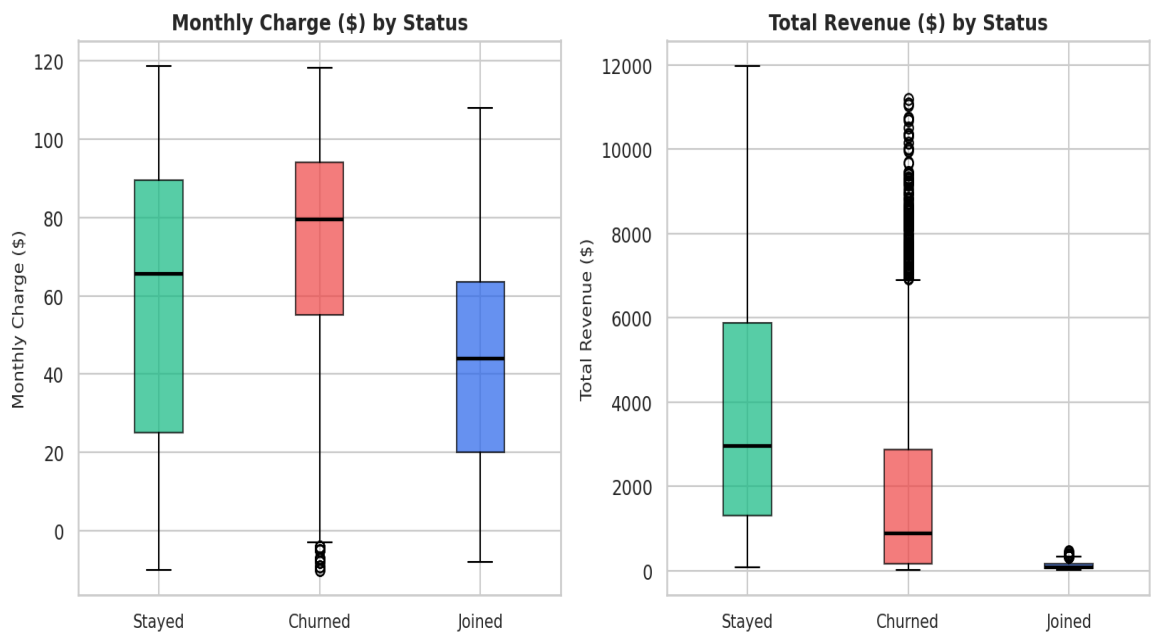


Figure 8: Monthly Charge & Total Revenue Boxplots by Status

6. Correlation Analysis

The heatmap below shows Pearson correlations between numeric features. Total_Charges and Total_Revenue are highly correlated (expected). Tenure is moderately positively correlated with Total_Charges.

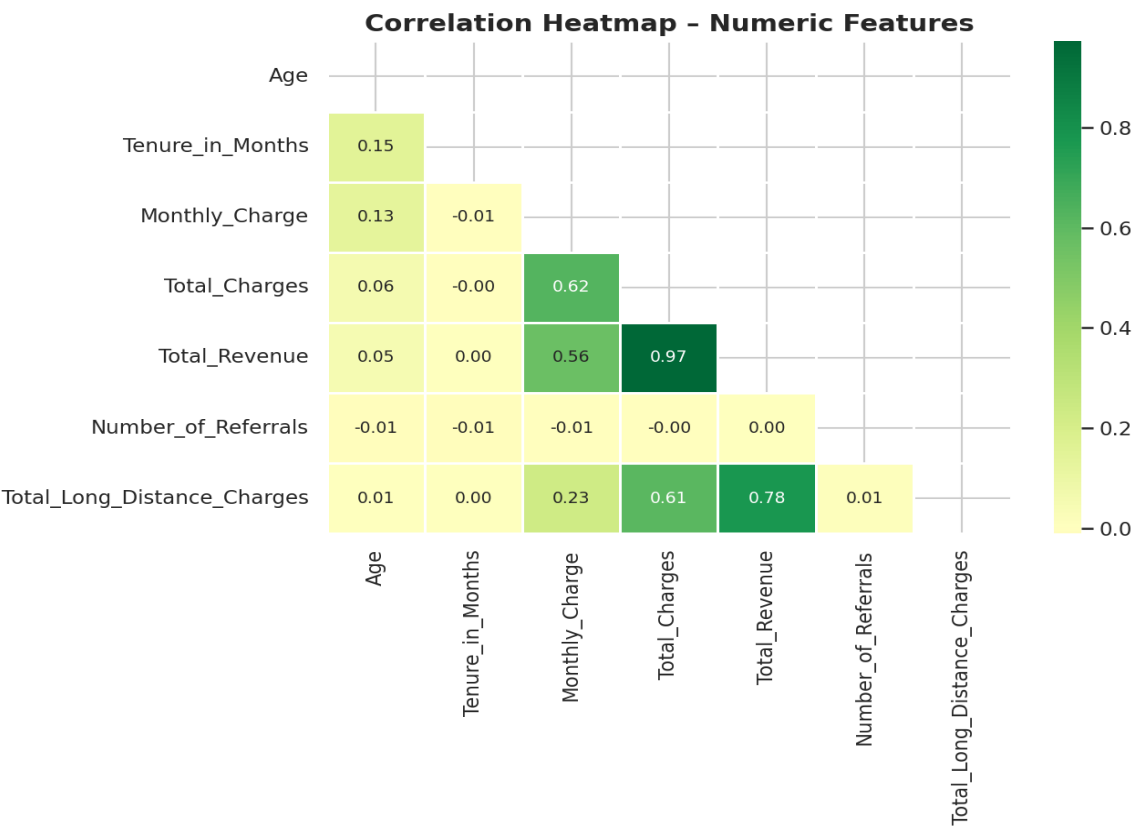


Figure 9: Correlation Heatmap

7. Geographic Analysis

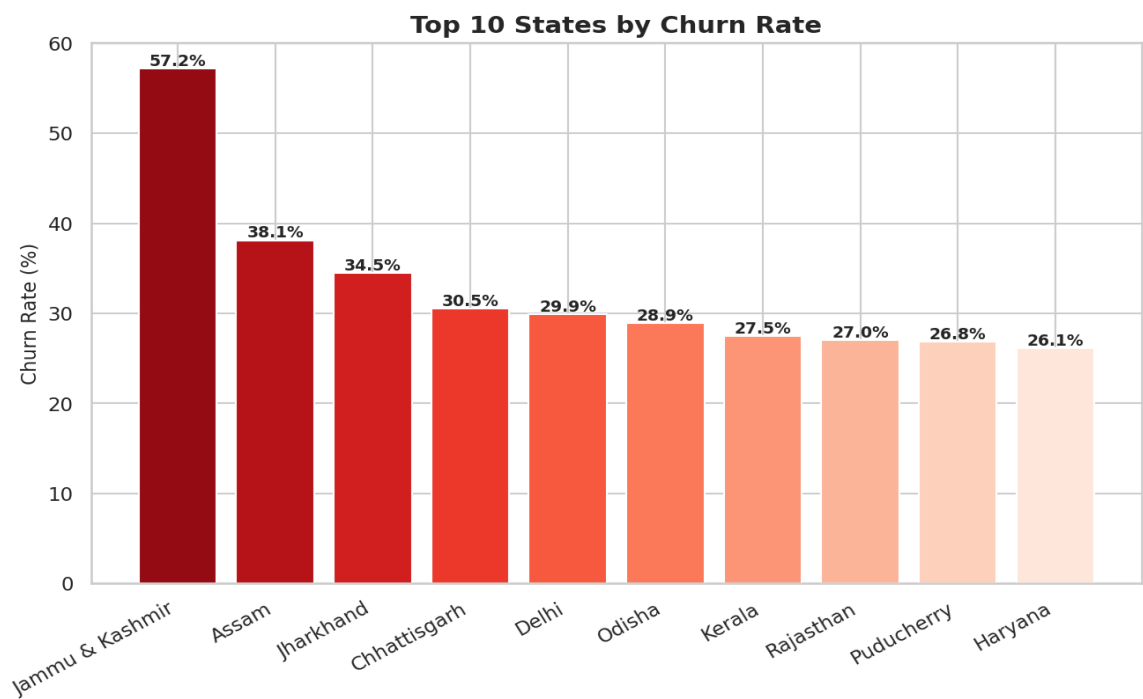


Figure 10: Top 10 States by Churn Rate

State	Churn Rate
Jammu & Kashmir	57.2%
Assam	38.1%
Jharkhand	34.5%
Chhattisgarh	30.5%
Delhi	29.9%

8. Key Findings & Recommendations

High Churn Rate

Overall churn is 27.0%, predominantly driven by competitor offers. Competitive pricing and service improvements are critical.

Month-to-Month Risk

Customers on Month-to-Month contracts churn at a significantly higher rate. Incentivizing migration to 1- or 2-year contracts can reduce churn.

Fiber Optic Churn

Fiber Optic internet users show higher churn, suggesting service quality or pricing issues that need targeted attention.

Revenue Impact

Churned customers had higher average monthly charges, meaning revenue loss per churned customer is above average.

Geographic Hotspots

Certain states show disproportionately high churn rates — targeted retention campaigns in those regions could deliver high ROI.

Tenure & Loyalty

Customers with longer tenure churn less. Early-tenure engagement programs (first 6 months) could significantly reduce early churn.